Experiment-1

Basic networking commands in linux and windows

Aim:

To study and execute basic networking commands available in linux and windows os for diagnosing and troubleshooting network issues

Requirements:

Pc with windows / linux, Terminal or command prompt Active network connection

Procedure:

1) open command prompt (windows) or terminal
2) Execute 'ipconfig' or 'ifconfig' to view ip configuration
3) use ping 8.8.8.8 to test connectivity to google DNS server
4) Execute 'tracer google.com' or 'traceroute google.com' to view path packet
5) Run 'netstat -an' to list active network connections and listening ports
6) use 'nslookup google.com' to perform DNS lookup

Commands.

1) ipconfig /all # windows full ip configuration

2) ip a # linux ip address info

3) ping -c 4 8.8.8.8 # linux ping 4 packets

4) ping 8.8.8.8 # windows ping

5) traceroute google.com # linux - trace route

6) netstat -tuln # linux - listen port

7) nslookup google.com # DNS resolution

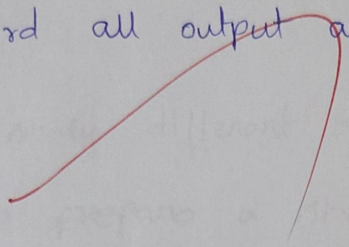
8) arp -a # Arp table

9) route -n # Routing table (linux)

7) Execute 'arp-a' to view the ARP cache

8) use 'route print' (windows) or 'route -n' (linux) to view the routing table

9) Record all output and analyse the results



Result:

Basic networking commands were successfully executed on both linux & windows. IP addresses, routing tables, ARP caches and DNS resolution were observed and recorded